

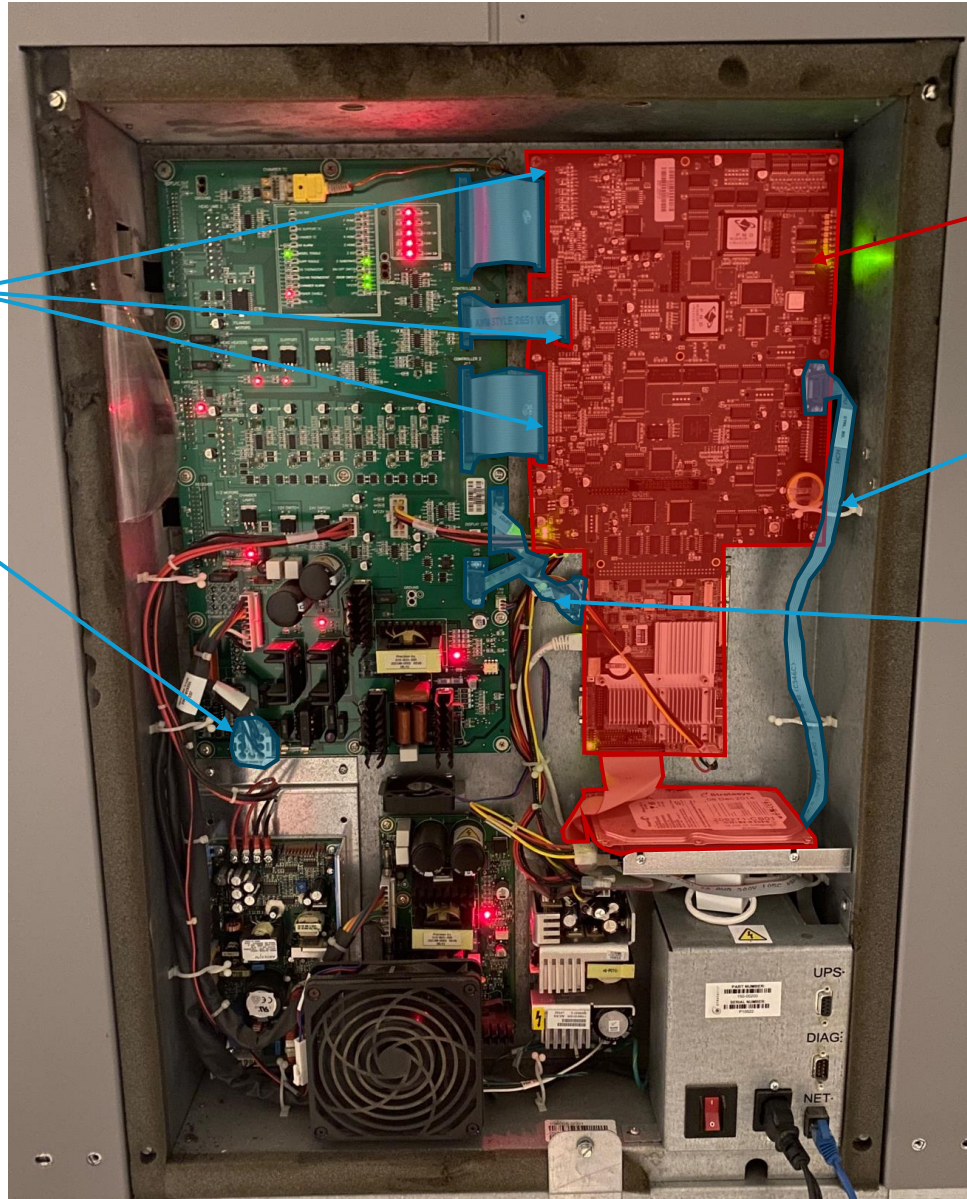
Disconnect 3x ribbon cables from the control board. Note the connectors are released via 2x lever locks on the short side of each board housing,

Disconnect power cable.

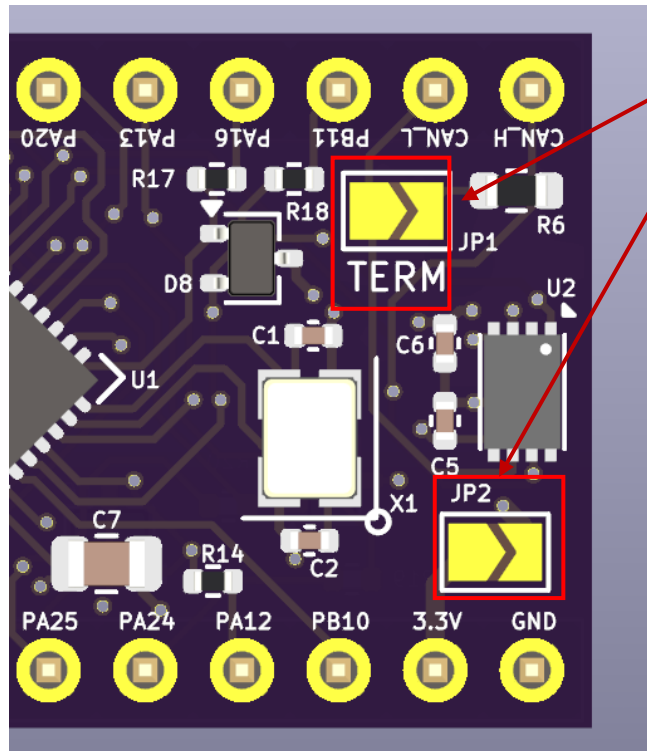
Remove and discard control board, SBC (single board computer), and hard drive. Discard screws and standoffs EXCEPT the screws used for the SBC, which are reused to mount the new 48V power supply.

Disconnect and tuck away/discard cable.

Disconnect and tuck away/discard cables.



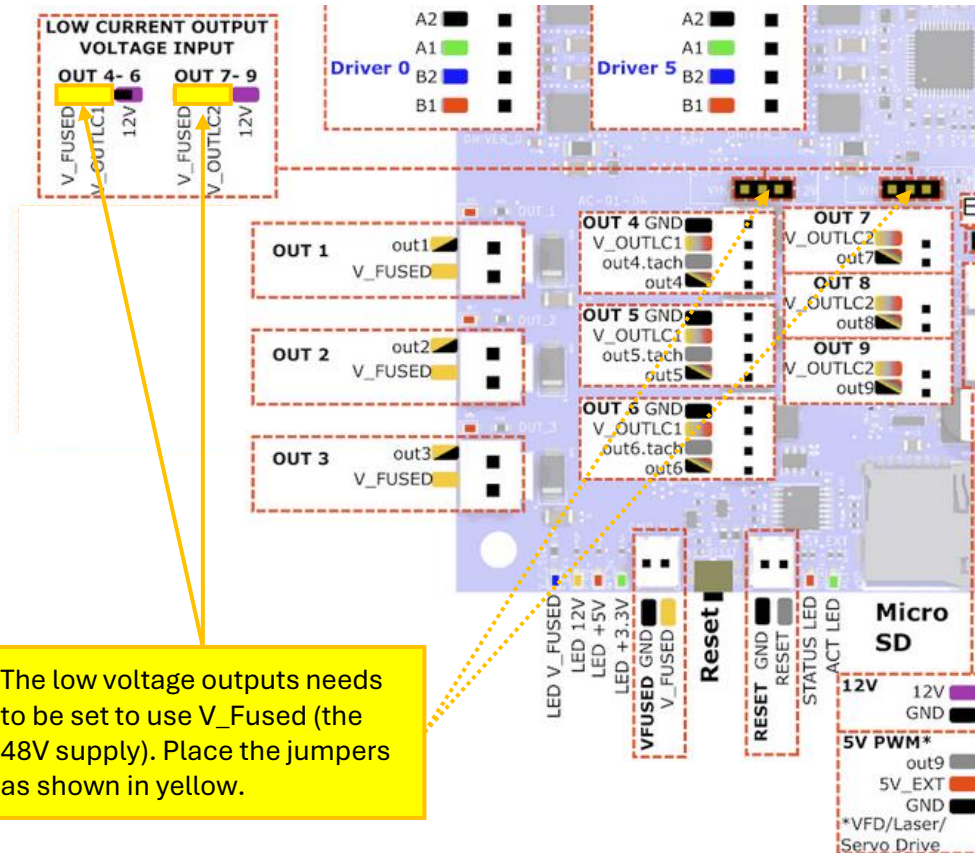
DP3EXB Prep



Leave JP1 (CAN Termination Resistor) and JP2 (3.3V to header pins) unbridged.

CAN termination is handled by the DuePrint3 interface board.

Duet 6HC Prep



The low voltage outputs needs to be set to use V_Fused (the 48V supply). Place the jumpers as shown in yellow.

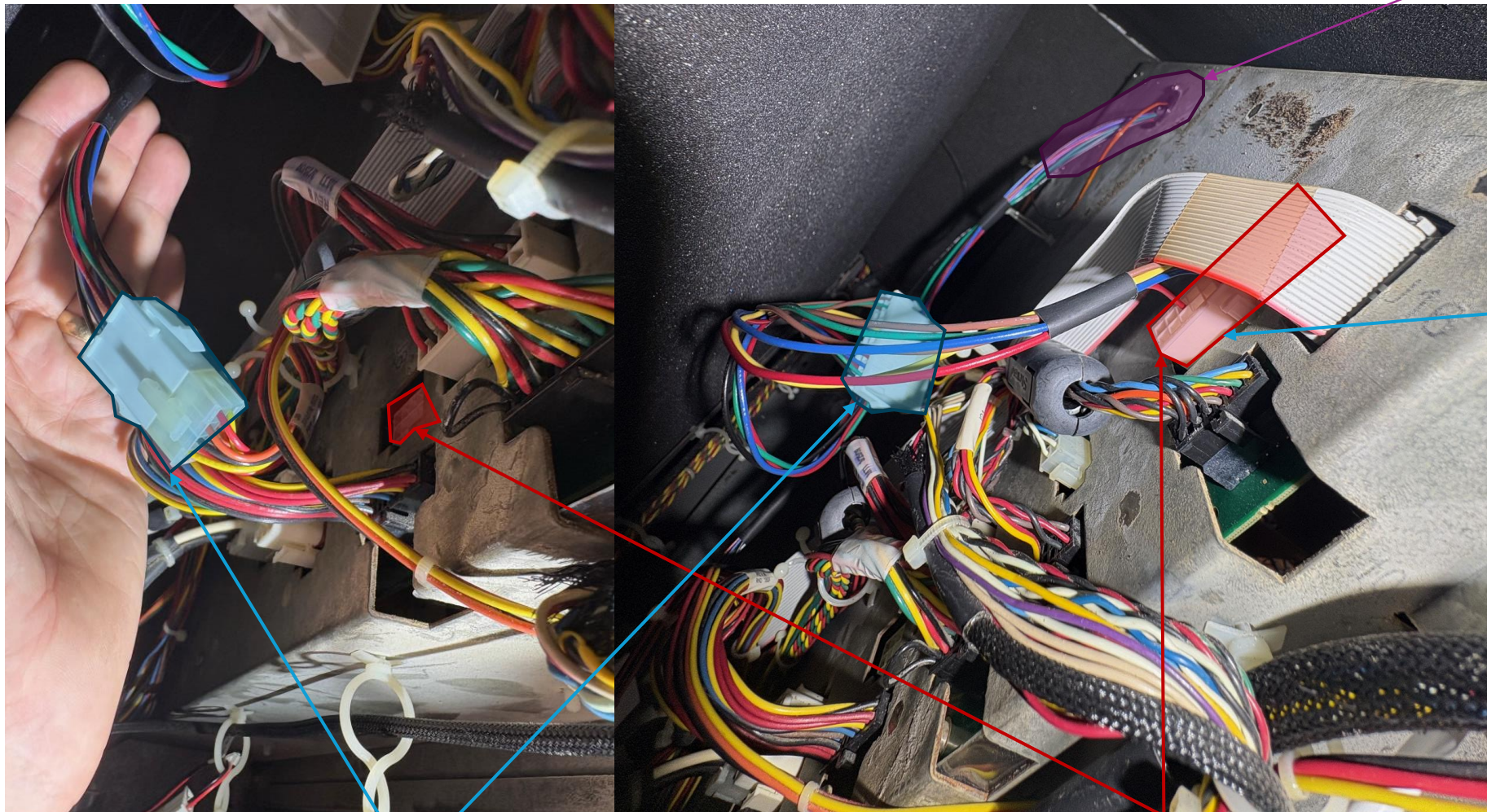
4. Pass the 3x Duet motor connectors (P4 P5 P6) through the feedthrough.

3. Connect the 20 pin Mini-fit Jr connector (P3) from the new interface cable to the PDB (J9).

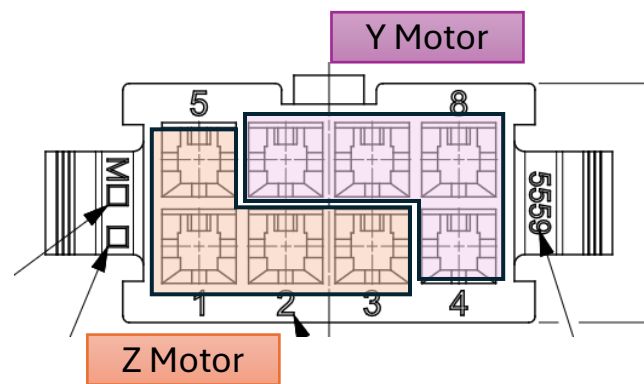
1. Disconnect the 8 pin and 20 pin Mini-Fit Jr connectors from the PDB (power delivery board).

The 8 pin board header will remain unused.

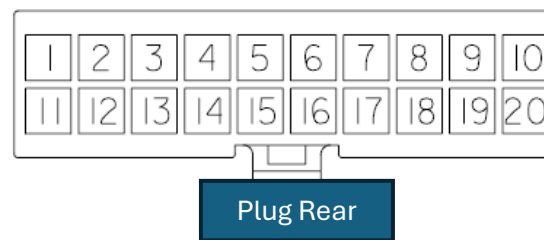
2. Connect the 8 pin (P2) and 20 pin (P1) Mini-Fit Jr connectors from the new interface cable to the existing wiring harness.



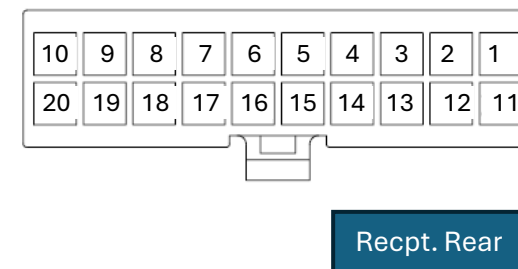
P2



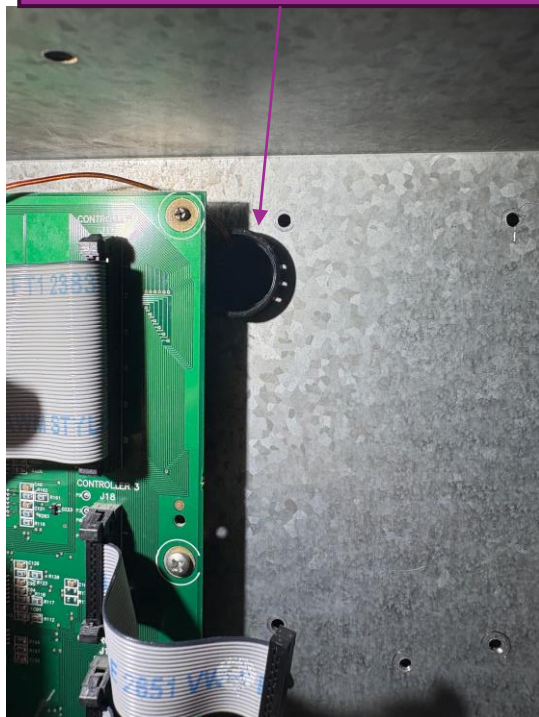
P3



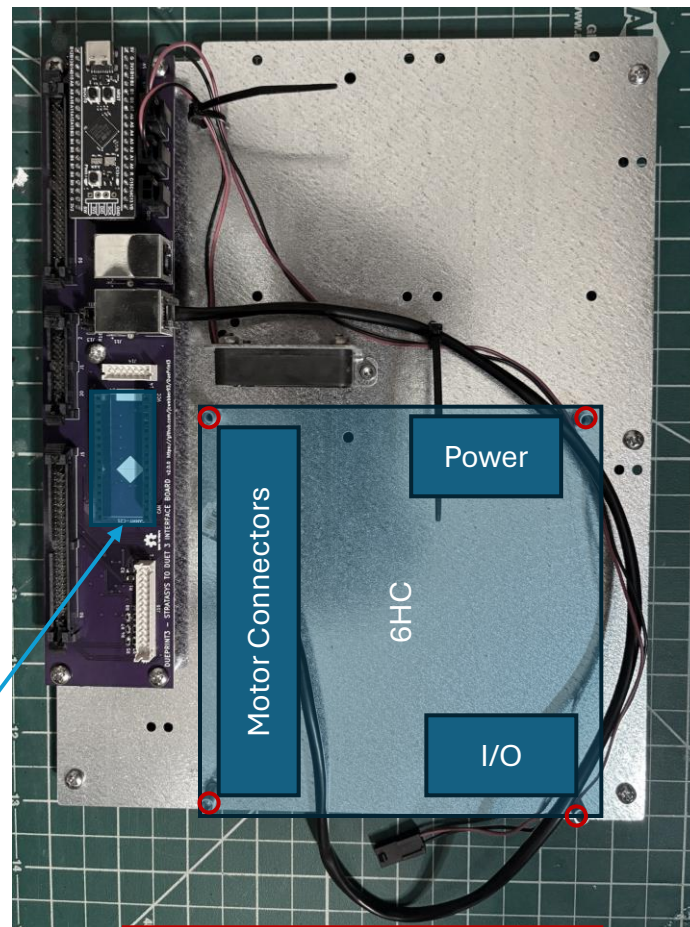
P1



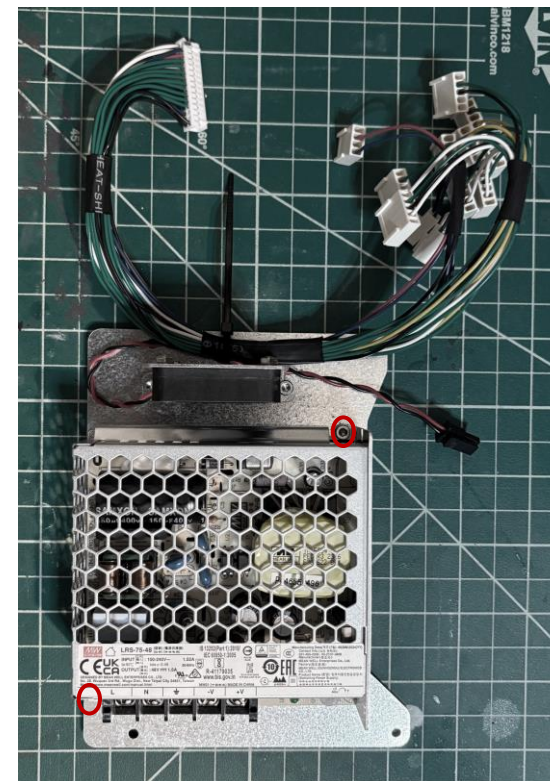
Pass the 3x Duet motor connectors through the feedthrough.



Install the DP3EXB as shown.



Mount the Duet 6HC to the panel with the 4x longer #6-32 screws.



Mount the 48V PSU to the power supply panel via the 2x M3 screws.

Connect the 3x ribbon cables to the DuePrint3 Board.

Connect JST connector to header on DuePrint3.

Connect stepper motor connectors.

Driver 2 = Z
Driver 1 = Y
Driver 0 = X

Attach the new electronics panel into the machine via the 6x shorter #6-32 screws (which are fed from the inside area where the other cables are routed).

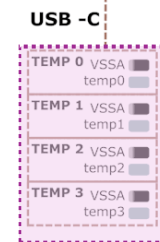
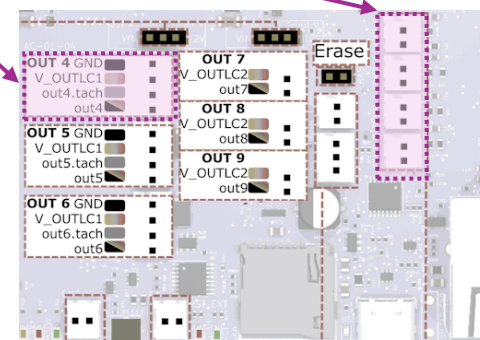
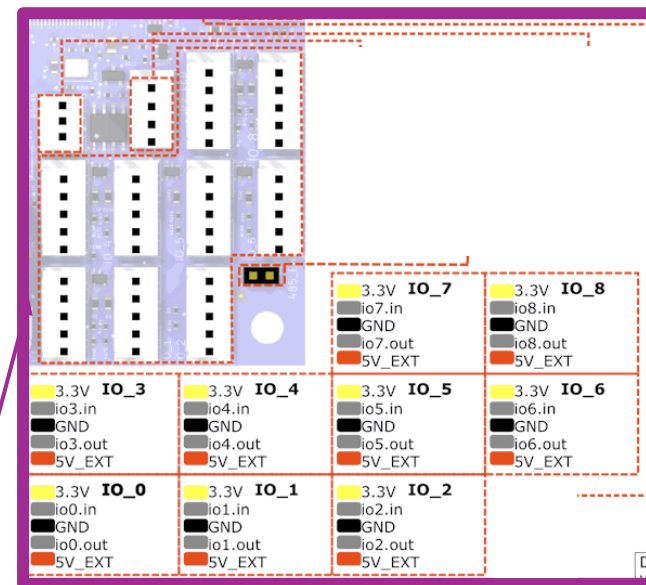
Connect RJ11 for CAN-FD

Connect I/O per markings on connectors

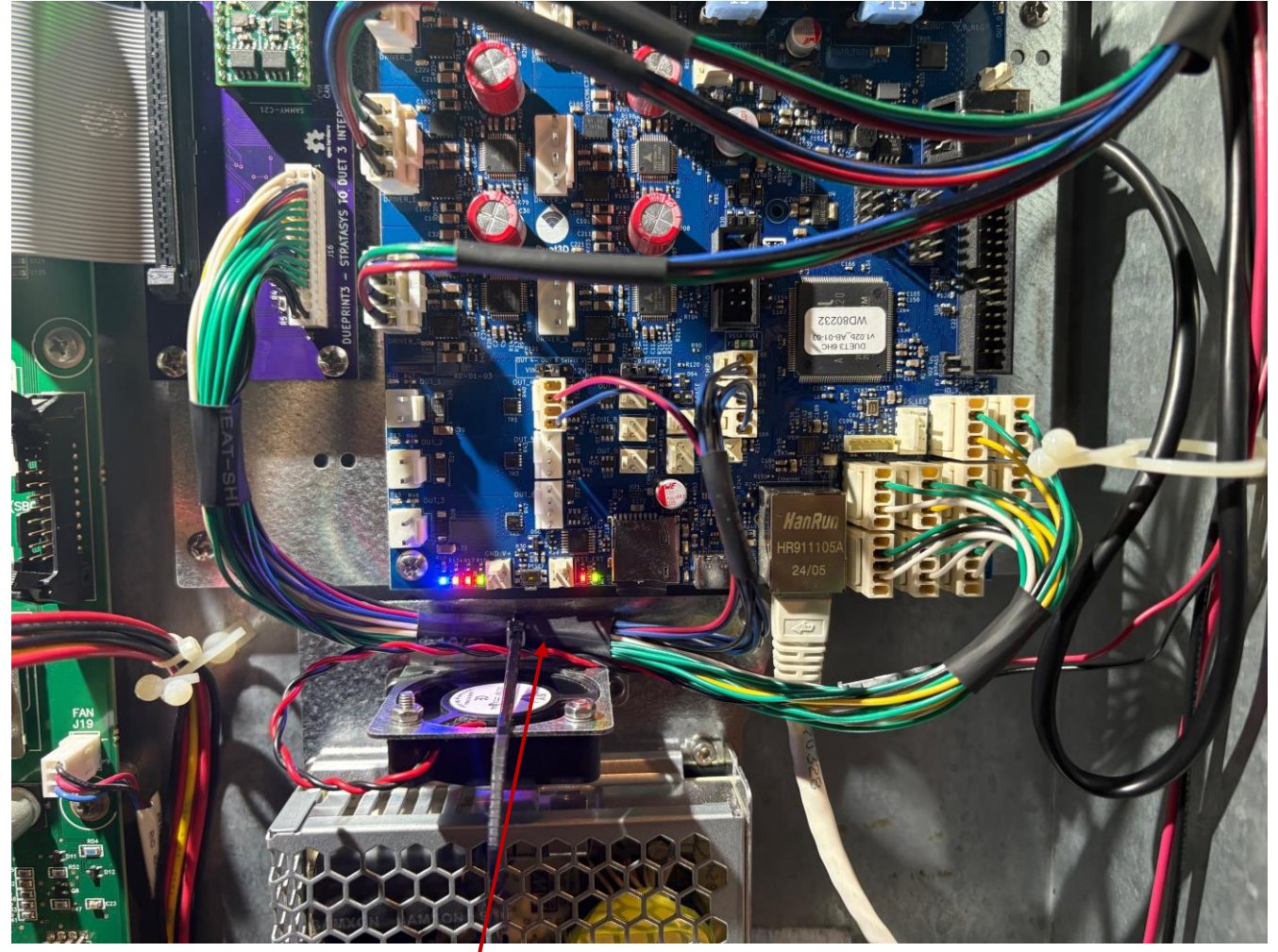
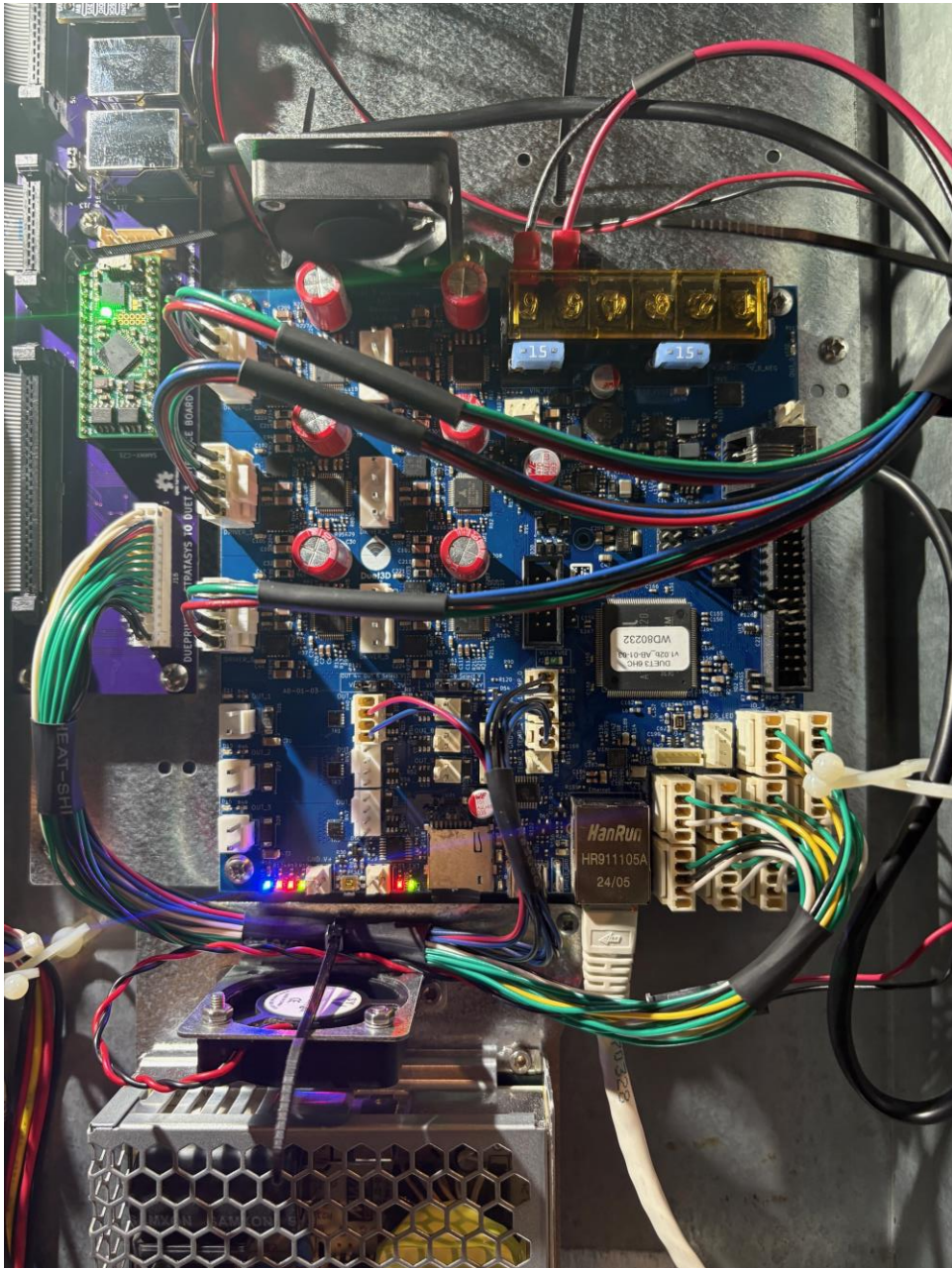
Plug ethernet cable into 6HC.

Connect 5V extension lead to 5V fan.

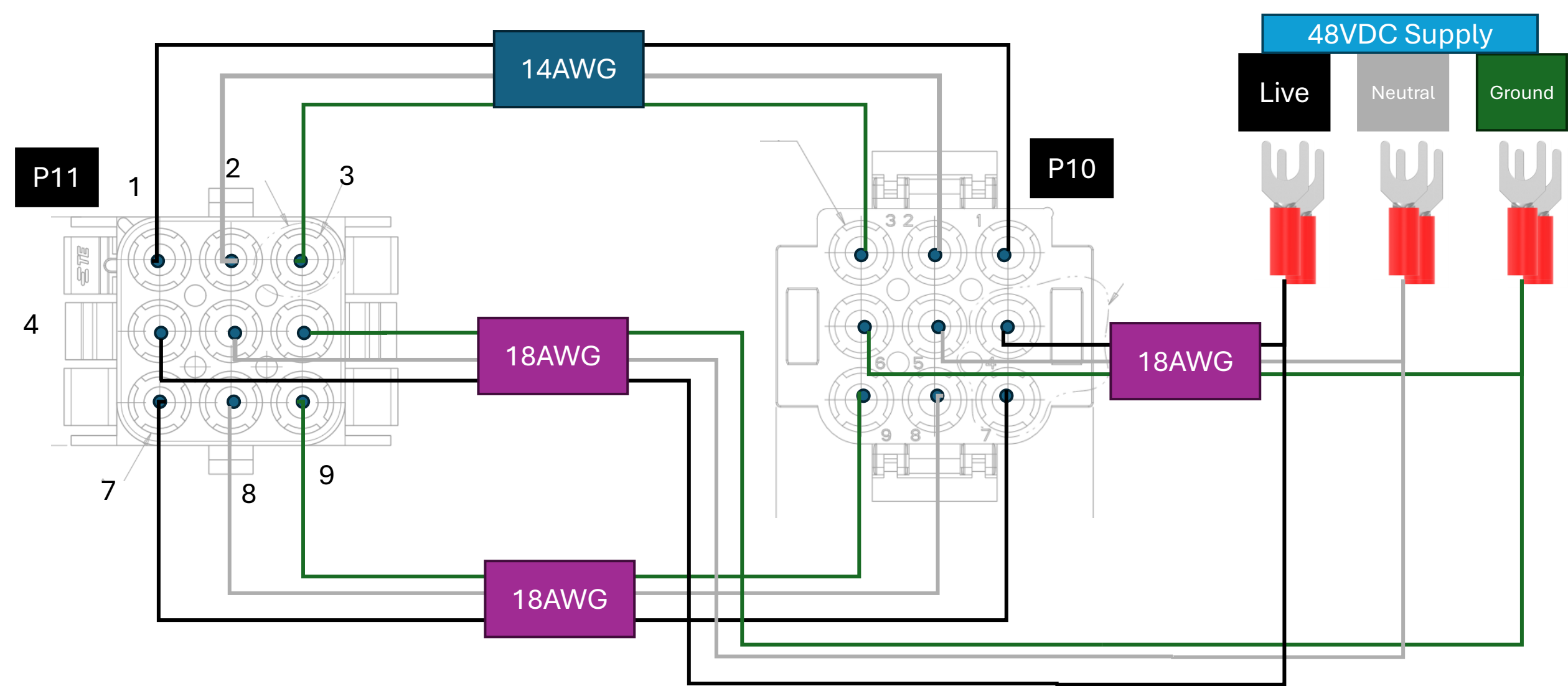
Attach the 48V power supply plate with the 4x screws previously used to mount the SBC.



Additional images of
connected Duet 6HC

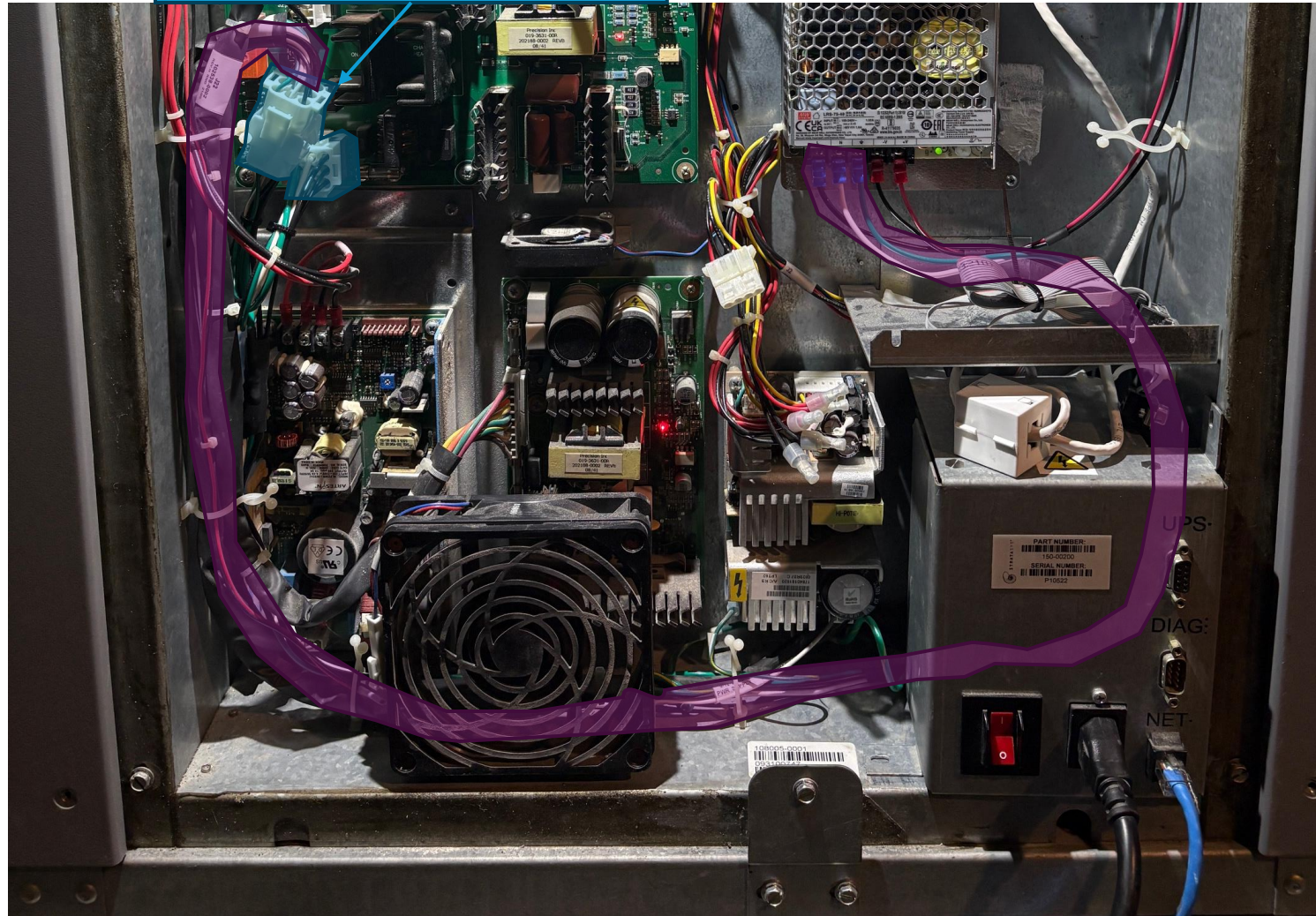


Ensure wires are not touching fan
blades.

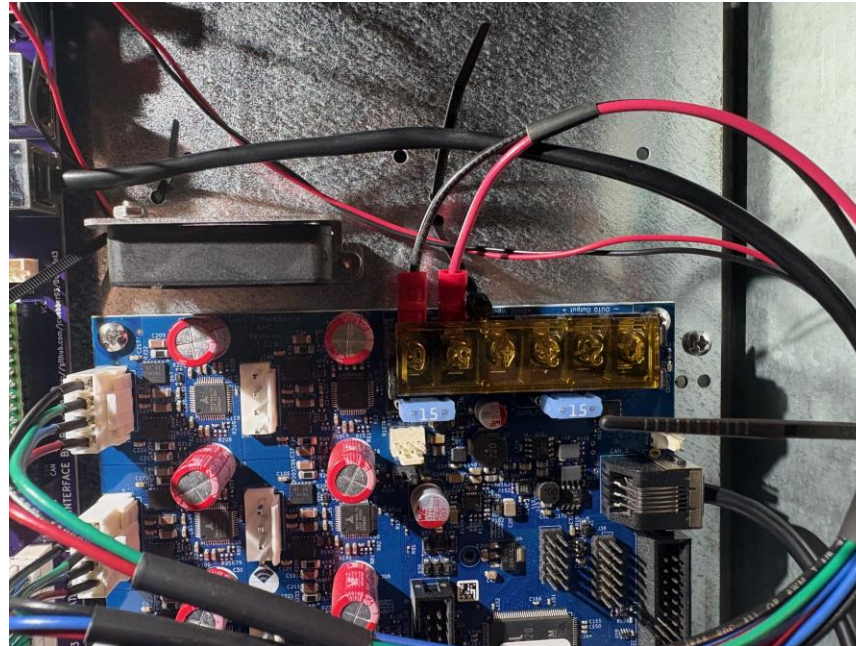


Example wiring for pins 4, 5, and 6. WAGO 221-413 or similar near P11 could be used instead of running VAC from P11 to the 48VDC supply, then back to P10

Connect new power harness to board.
Connect original power plug to new power harness.



Route power leads approximately as shown. You may wish to temporarily remove the power inlet module, which is retained by 3x screws.



Attach power terminals per the 6HC documents (for 48V) and 48V and AC power per your 48v power supply documents.

AC

Live

Neutral

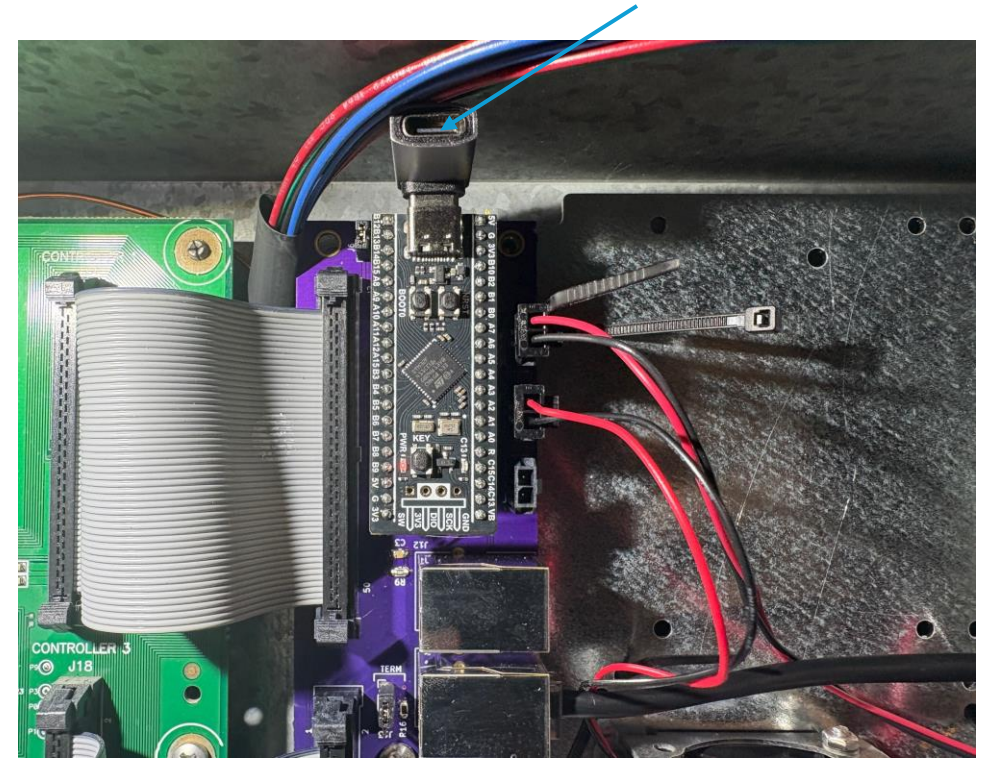
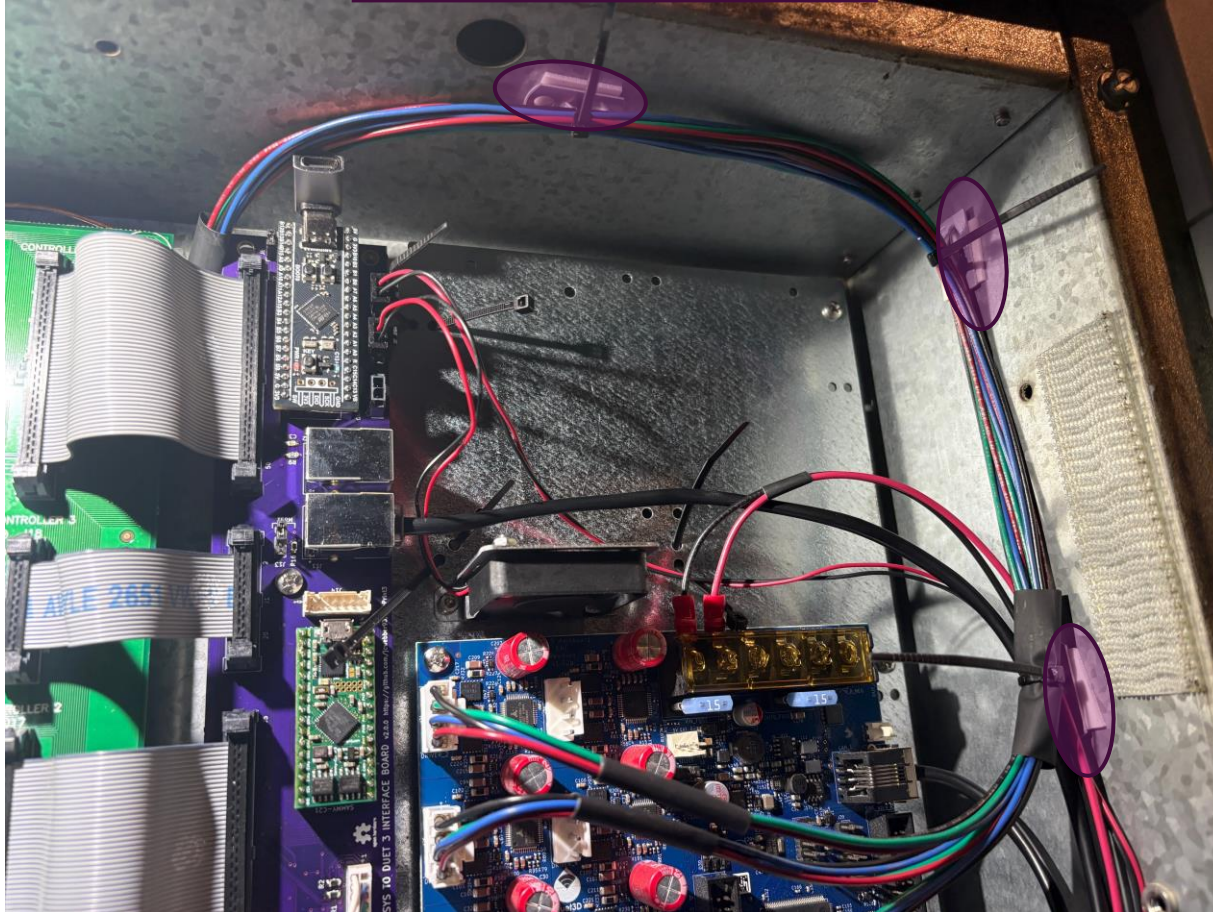
Ground

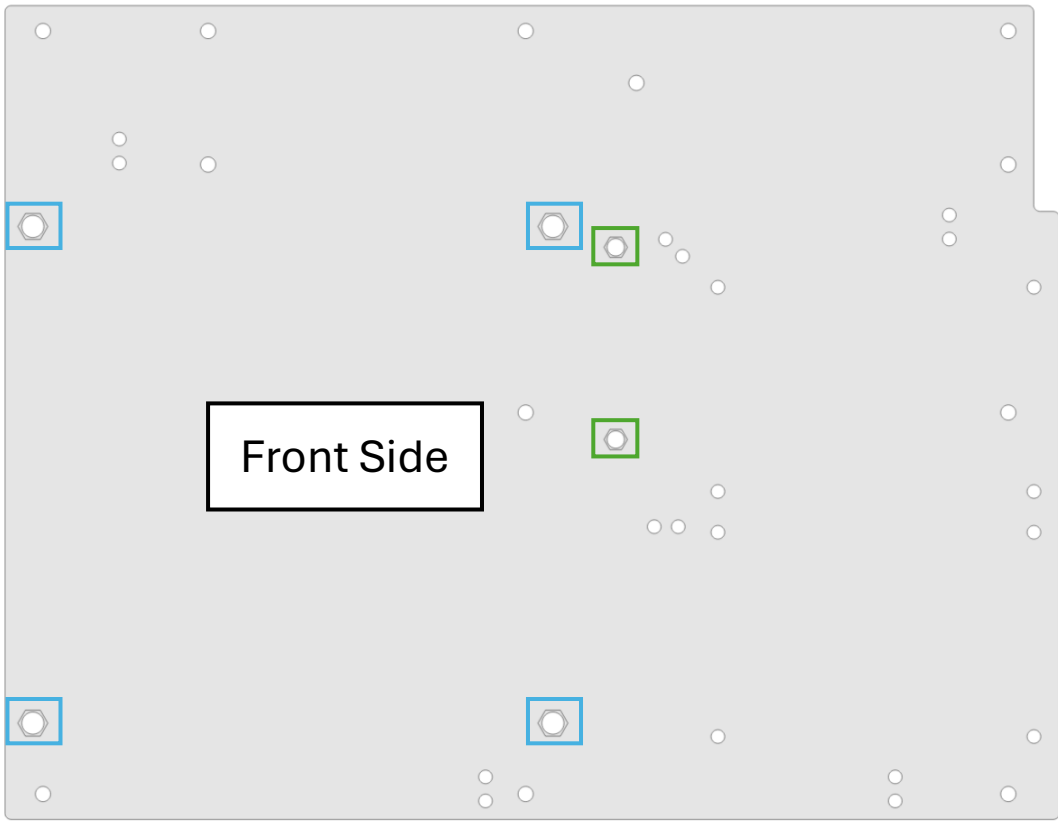
48VDC

48V

Ground

May wish to add tie down plates to aid with cable management.





6 Hardware Insertions

One feature per hole

Diameter	Feature
> ☒ 0.2130"	FLUSH STANDOFF, 6-32, .500" ⓘ ⓘ ↑ ↓
> ☒ 0.1662"	NUT, M3X0.5 ⓘ ⓘ ↑ ↓
> 0.1494"	OPEN CATALOG
> 0.1338"	OPEN CATALOG

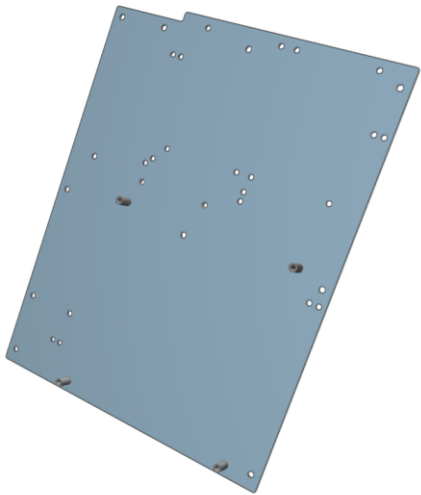
RESET ALL

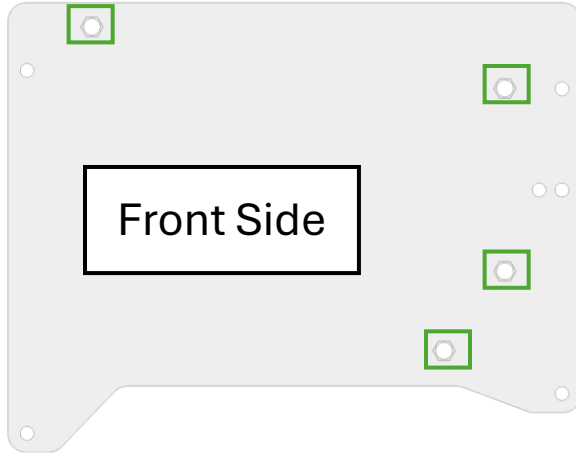
DONE

Studs on Front Side
Nuts on Back Side

0.048" G90 Galvanized Steel

Interface Plate.STEP





4 Hardware Insertions
One feature per hole

Diameter	Feature
0.1662"	NUT, M3X0.5 ⓘ ⊗
> 0.1406"	OPEN CATALOG

RESET ALL DONE

Nuts on Back Side

0.048" G90 Galvanized Steel

48VDC Plate.STEP

